



National Teachers College
629 J Nepomuceno, Quiapo, Manila, 1001 Metro Manila
Bachelor of Science in Information Technology

Final Project/Exam: System Analysis and Design (SAD) for Sustainable Development

System Analysis and Design

Submitted By:

Aboc, Melchor

Arongo, Jalil

Nalda, Jan Jb

Natan, Jazy Anne

Sison, Heaven Romeo

Terrado, Al Cedric

Block and section:

BSIT - 2.3

Submitted To:

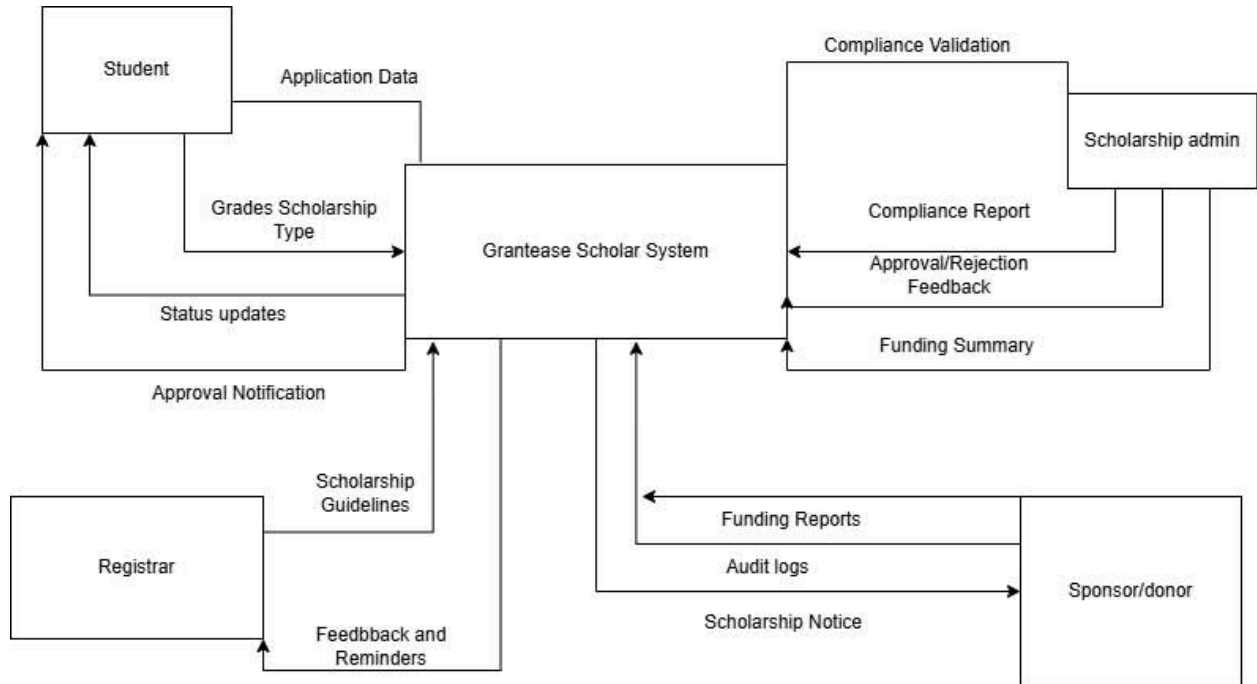
Ms. Justin Louise R. Neypes



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Software Design & Analysis Document (SDAD)

Process Architecture

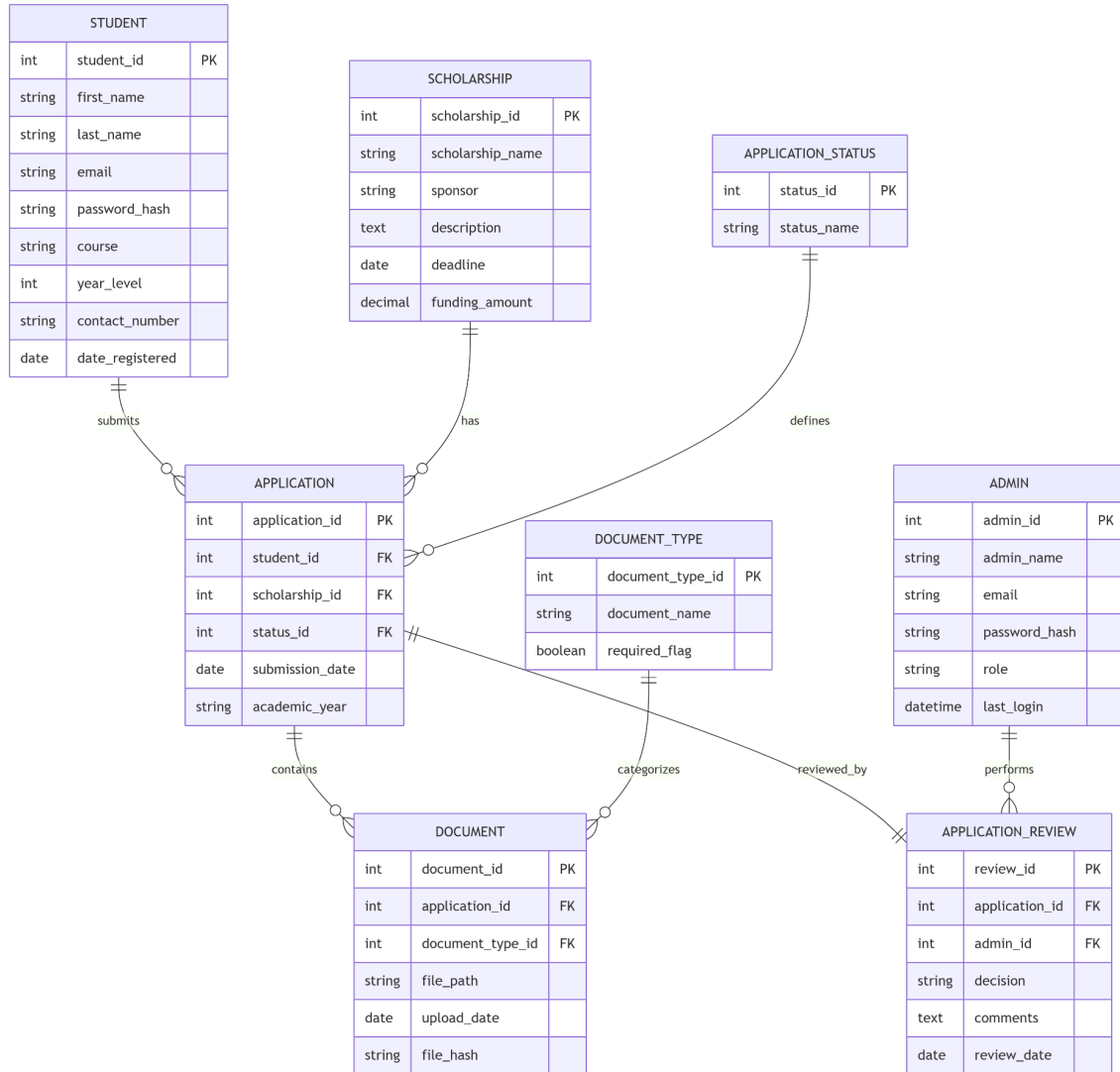




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System Infrastructure





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TELOS Feasibility

Dimension	Question	Our Answer	Risk Level
Technical	Do we have the skills and tools?	Yes – PostgreSQL, Node.js, React are all free and we know them	Low
Economic	Can we afford it?	Yes – development costs near zero; hosting is cheap (P0-P5,000)	Low
Legal	Are we following the law?	Yes – but we need to be careful with RA 10173; lawyer will review	Medium
Operational	Will people actually use it?	Yes – Registrar and Admin roles already exist; just need 2 days of training	Low
Scheduling	Can we finish on time?	Yes – 14 weeks is enough for the must-have features	Medium



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UI/UX BLUEPRINT

GRANTEASE – SCHOLARSHIP MANAGEMENT SYSTEM | BLUEPRINT

RESPONSIVE BREAKPOINTS: 375px (Mobile) 768px (Tablet) 1440px (Desktop)

1 LOGIN PAGE



GrantEase

Scholarship Management System

Student

Admin

Registrar

Sponsor

Sign In →

Forgot Password?

2 STUDENT DASHBOARD

Welcome back, Maria Santos!

Track your applications, view your GPA

Total Applications: 3

Under Review: 2

Approved: 1

Current GPA: 1.5 (1.0 best)

+ Apply Now

Track Application

Approval Status

My Applications

NTC Excellence Scholarship

Submitted: Mar 15, 2025

Amount: P20,000

Tracking: GRT-2025-0001

GPA Required: 1.5

Submitted

Review

Decision

View Details

Dean's List Grant

Submitted: Mar 28, 2025

Amount: P15,000

Tracking: GRT-2025-0002

GPA Required: 1.2

Submitted

Review

Decision

View Details

Apply early! 40% more likely to be reviewed

Clear & Legible documents increase approval chance

3 APPLICATION FORM

Apply for Scholarship

Your GPA: 1.5

Select Scholarship

-- Select Scholarship --

First Name: Maria

Last Name: Santos

Student ID: NTC-2024-01234

Email: maria@ntc.edu.ph

Course: -- Select Course --

Personal Statement: Write your personal statement here...

Click to upload documents (PDF, JPG, PNG max 10MB)

Cancel

Submit Application

4 TRACK APPLICATION

Track Your Application

Enter your tracking ID

GRT-2025-0001

Track Now

NTC Excellence Scholarship

Status: Under Review

Submitted: Mar 15, 2025

Submitted

Compliance

Decision

Recent Applications

Dean's List Grant

GRT-2025-0002

Approved

Track This

Financial Aid

GRT-2025-0003

Under Review

Track This

NTC Excellence

GRT-2025-0001

Pending

Track This

Need Help?

scholarship@ntc.edu.ph

(02) 1234 5678

5 ADMIN DASHBOARD

Welcome back, Admin!

Review pending applications

187 Total Applicants +12%

2 Pending Review Need attention

94 Approved This semester

P1.85M Total Disbursed 5 scholarships

Pending (2)

Approved (94)

Students

Reports

Settings

Student	Scholarship	Submitted	GPA	Documents	Actions
Juan Dela Cruz	NTC Excellence	Mar 10, 2025	1.5	Complete	✓ ✗
Maria Santos	Dean's List	Mar 12, 2025	1.2	Pending	✓ ✗

✓ Approve

⌚ Review

✗ Reject

6 REGISTRAR DASHBOARD

Welcome back, Registrar!

Review compliance queue

3 Pending Verification

4 Verified Records

Compliance Queue

Verified Records

Student	Scholarship	Submitted	Required Documents	Action
Juan Dela Cruz	NTC Excellence	Mar 10, 2025	Grades Certificate	✓ ✗
Jose Ribal	Financial Aid	Mar 14, 2025	Income Statement Grades	✓ ✗

Student	Scholarship	Verified Date	Status	Verified By
Maria Santos	Dean's List	Mar 1, 2025	Compliant	Registrar Cruz

7 SPONSOR DASHBOARD

Welcome, Scholarship Sponsor!

Track your funding impact

P1,245,000 Total Fund Disbursed

89 Scholars Supported

5 Active Scholarships

78% Utilization Rate

Funding Summary

Allocated: P1,600,000

Disbursed: P1,245,000

Download Report

Impact Metrics

Avg. GPA: 1.6

Retention Rate: 98%

Graduation Rate: 94%

Refresh

Upcoming Disbursements

May 15, 2025: P250,000

June 1, 2025: P180,000

Transparency Overview

Scholarship	Total	Disbursed	Remaining	Scholars	Utilization
NTC Excellence	P500,000	P400,000	P100,000	20	80%
Dean's List	P450,000	P380,000	P70,000	25	84%
Financial Aid	P600,000	P465,000	P135,000	18	78%

Generate Full Transparency Report

8 APPROVAL STATUS PAGE

Approval Status

NTC Excellence Scholarship

Submitted: Mar 15, 2025

In Progress

Pending

Decision

Your application is being reviewed. Check back soon!

Dean's List Grant

Submitted: Feb 28, 2025

Completed

Approved!

Decision

Congratulations! Your scholarship has been approved.

Financial Aid Program

Submitted: Mar 20, 2025

In Progress

Pending

Decision

Your application is under review. We'll update you soon!



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Validation Plan

Traceability Matrix

Requirement	What It Does	Test Case	How We'll Test It
FR-01	Digital application submission	TC-01	Student submits form with attachments – does it save correctly?
FR-02	Confirmation within 60 seconds	TC-02	Measure time from submit to email arrival
FR-03	Real-time status tracking	TC-03	Change status; does the student see it immediately?
FR-04	Registrar compliance validation	TC-04	Registrar marks compliant/non-compliant – does it save?



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FR-05	Scholarship Admin approval	TC-05	Admin approves/rejects – does final status update?
FR-06	Audit log maintenance	TC-06	Access data – does log show who, when, what?
NFR-01	99.9% uptime	TC-07	Simulate 500 users at once – does the system stay up?
NFR-03	256-bit encryption	TC-08	Check database – are PII fields encrypted?
FR-08	Role separation	TC-09	Registrar tries Admin functions; Admin tries Registrar functions – both should fail